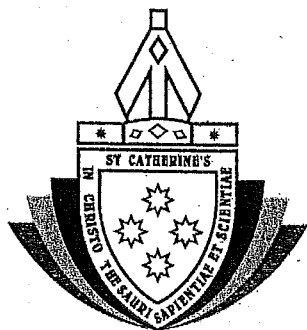




Student Number: _____

St. Catherine's School

Waverley

August 2011

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Complete each question in a separate booklet

- **Total Marks – 120**
- **Attempt Questions 1 – 10**
- **All questions are of equal value**

Total marks - 120

Attempt Questions 1-10

All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 1 (12 marks) Use the Question 1 Writing Booklet.

- (a) Write down the equation of a circle with centre $(3, -4)$ and radius 5 units. 1
- (b) If $a = \frac{1}{2}$ and $b = -\frac{1}{3}$, find the value of $\frac{a+b}{a-b}$. 1
- (c) Solve $8x^2 = 2x$. 2
- (d) Express $\frac{1}{\sqrt{3}} + \frac{1}{2+\sqrt{3}}$ with a rational denominator. 2
- (e) Find $\sum_{n=1}^{100} (4n+3)$ 2
- (f) Find $\lim_{x \rightarrow 2} \frac{x-2}{x^3-8}$. 2
- (g) If the limiting sum of a geometric series is 48 and the first term is 4, find the common ratio, r . 2

Question 2 (12 marks) Use the Question 2 Writing Booklet.

(a) Solve $|2x - 3| > 5$ and graph your solution on a number line.

3

(b) Show that $\frac{d}{dx} x e^{\sin x} = e^{\sin x} (1 + x \cos x)$.

1

(c) Find $\int \frac{1}{(5x-7)^3} dx$.

1

(d) Evaluate:

(i) $\int_0^1 e^{\pi x} dx$

2

(ii) $\int_{-1}^0 \frac{dx}{2x+3}$

2

(e) Let $f(x) = \log_e(x-2)$. What is the domain of $f(x)$?

1

(f) Find the gradient of the tangent to the curve $y = e^{\frac{x}{2}}$ at the point where $x = 2$.

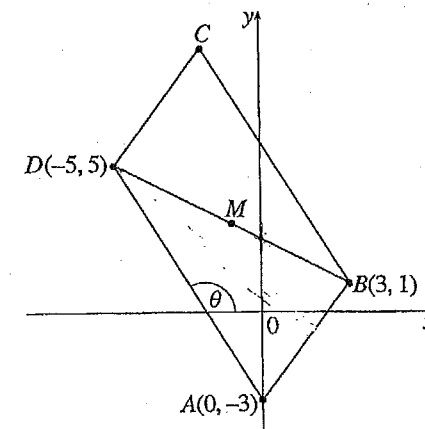
2

Question 3 (12 marks) Use the Question 3 Writing Booklet.

(a) How many sides has a regular polygon if each of its internal angles is 168° ?

2

(b)



Not to scale

$A(0, -3)$, $B(3, 1)$, $C(x, y)$ and $D(-5, 5)$ are the vertices of a parallelogram, as shown in the diagram above.

(i) Find θ to the nearest degree.

2

(ii) Find the coordinates of M , the midpoint of BD .

1

(iii) Find the coordinates of C .

1

(iv) Show that the line AB has equation $4x - 3y - 9 = 0$.

2

(v) Find the perpendicular distance between D and the line AB .

2

(vi) Find the area of the parallelogram $ABCD$.

2

Question 4 (12 marks) Use the Question 4 Writing Booklet.

- (a) (i) The fourth term of a geometric series is -27 and the seventh term is 729 . Find the first term and common ratio. 2
- (ii) Find the sum of the first seven terms of the series. 2

- (b) (i) Copy and complete the table below. Give your answers to 2 decimal places. 1

x	1	2	3	4	5
$\frac{4}{x(x+1)}$					

- (ii) Use all the information above and Simpson's rule, to find

an approximation to $\int_1^5 \frac{4}{x(x+1)} dx$. 2

- (iii) Show that $\frac{4}{x} - \frac{4}{x+1} = \frac{4}{x(x+1)}$. 1

- (iv) Deduce the value of the integral $\int_1^5 \frac{4}{x(x+1)} dx$. 2

- (v) Hence find the percentage error given by Simpson's rule. 2

Answer to 2 decimal places.

Question 5 (12 marks) Use the Question 5 Writing Booklet.

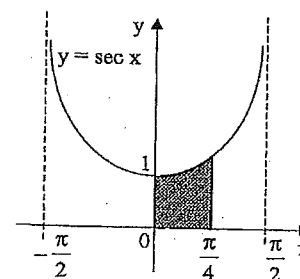
- (a) P is the parabola $y^2 + 4y = x + 3$.

- (i) Express P in the form $(y - k)^2 = 4a(x - h)$. 2

- (ii) Find the equation of the directrix. 1

- (iii) Find the coordinates of the focus. 1

- (b)



Not to scale

The shaded region which lies between the x axis and the curve $y = \sec x$

from $x = 0$ to $x = \frac{\pi}{4}$ is rotated about the x axis to form a solid.

Find the volume of the solid. 3

- (c) A pool is being drained and the number of litres of water, L , in the pool at time t minutes is given by the equation $L = 120(40 - t)^2$.

- (i) At what rate is the water draining out of the pool when $t = 6$ minutes? 2

- (ii) How long will it take for the pool to completely empty? 1

- (d) Consider the function whose derivative is given by $\frac{dy}{dx} = x^3(x-2)(x+3)$. 2

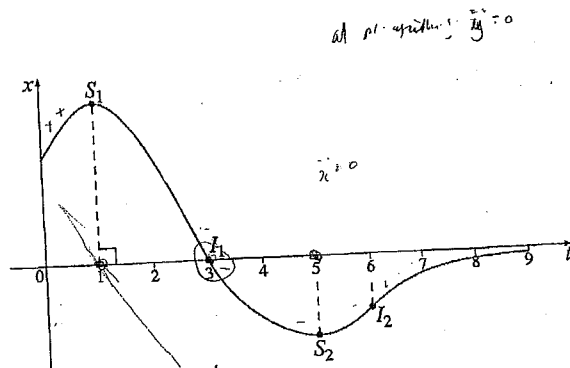
At what value of x will a maximum turning point exist on the graph of the function? Give a reason for your answer.

Question 6 (12 marks) Use the Question 6 Writing Booklet.

(a) Solve the equation $e^{4x} + 2e^{2x} = 8$.

3

(b)



The graph shows the position of a particle, moving on a straight line, for the first nine seconds of the motion. S_1 and S_2 are stationary points, I_1 and I_2 are points of inflexion.

State the times, or periods of time, for which:

- (i) the particle is stationary
- (ii) the velocity is negative
- (iii) the acceleration is positive

1

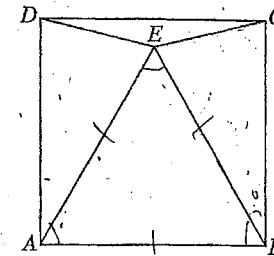
1

1

Question 6 continues on page 9

Question 6 (continued)

(c)



$ABCD$ is a square and $\triangle ABE$ is an equilateral triangle.

(i) Prove $\angle EBC = 30^\circ$.

1

(ii) Prove triangles EBC and EAD are congruent.

2

(iii) If the lengths of the sides of the square are d cm, prove that the area of triangle CDE is $\frac{d^2(2-\sqrt{3})}{4} \text{ cm}^2$.

3

End of Question 6

Question 7 (12 marks) Use the Question 7 Writing Booklet.

- (a) A particle is moving in straight line so that at time t seconds its displacement from the origin is x metres. Initially the particle is 1 metre to the left of the origin.

The velocity of the particle is given by $v = 2\cos t - 1$.

- Express the displacement x as a function of t .
- At what time is the particle first at rest?
- Find the position of the particle at this instant.
- Draw the graph of $v = 2\cos t - 1$ for $0 \leq t \leq 2\pi$, showing clearly all intercepts with the axes.

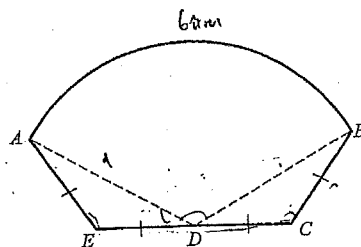
2

2

1

2

- (b) The Smiths are building an unusually shaped pool in their backyard.



In the diagram, $ABCDE$ represents the shape of the surface of the pool.

The sector ABD has centre D and $\angle ADB = \frac{2\pi}{3}$. The points C , D and E

lie on a straight line. The arc AB has a length of 6π metres. $AE = ED = DC = CB$.

- Show that $AD = 9$ metres.
- Find the exact length of BC .
- Find the exact area of the pool's surface.

1

2

2

Question 8 (12 marks) Use the Question 8 Writing Booklet.

- (a) For what values of k does the quadratic equation $(5k - 3)x^2 - 4kx + k + 1 = 0$ have no real roots?

3

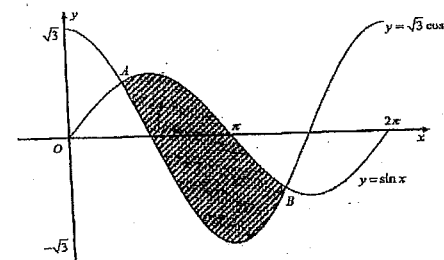
- (b) In a netball competition between the Green team and the Blue team, on average the Green team has won three out of the four of their games.

- Find the probability that the Green team wins the next two games.
- In the next three games, what is the probability that the Green team wins more games than the Blue team?

2

2

- (c) The diagram shows the graph of $y = \sin x$ and $y = \sqrt{3} \cos x$, $0 \leq x \leq 2\pi$. The graphs intersect at points A and B .



Not to scale

- Show that point A has co-ordinates $\left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right)$.
- Find the coordinates of point B .
- Find the shaded area enclosed by the two graphs.

1

1

3

Question 9 (12 marks) Use the Question 9 Writing Booklet.

- (a) Mark borrows \$20 000 from the ABC bank. This loan plus interest is to be repaid in equal monthly instalments of \$399 over five years. Interest of 7.2% p.a is compounded monthly on the balance owing at the start of each month.

Let A_n be the amount owing after n months.

- (i) Over the five year repayment period, how much interest is charged? 1
- (ii) Show that $A_1 = 19721$. 1
- (iii) Clearly show that $A_2 = 20000 \times 1.006^2 - 399(1 + 1.006)$. 2
- (iv) Deduce then that $A_n = 66500 - 46500 \times 1.006^n$. 2
- (v) After two years of repayments Mark decides on the very next day to repay the loan in one full payment. How much will this one payment be? 1

- (b) The amount Q grams of a carbon isotope in a dead tree trunk is given by $Q = Q_0 e^{-kt}$ where Q_0 and k are positive constants and time t is measured in years from the death of the tree.

- (i) Show that Q satisfies the equation $\frac{dQ}{dt} = -kQ$. 1
- (ii) Show that if the half life of the carbon isotope is 5500 years (i.e. if it takes 5500 years for the carbon isotope to reduce to half its mass), then $k = \frac{\ln 2}{5500}$. 2
- (iii) For a particular dead tree trunk, the amount of carbon isotope is only 15% of the original amount in the living tree. 2

How long ago did the tree die? Give your answer to the nearest 1000 years.

Question 10 (12 marks) Use the Question 10 Writing Booklet.

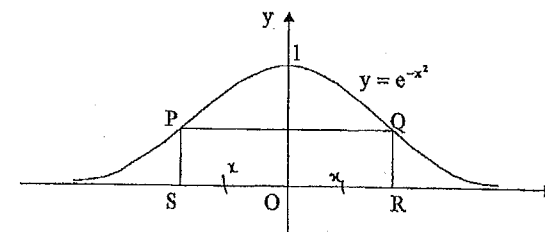
- (a) The sum of the first 10 terms of the series,

$$\log_2\left(\frac{1}{x}\right) + \log_2\left(\frac{1}{x^2}\right) + \log_2\left(\frac{1}{x^3}\right) + \dots (x > 0) \text{ is } 440.$$

Find the value of x .

3

- (b) The diagram shows a rectangle $PQRS$ where P and Q are on the curve and R and S are on the x axis. The point O is the origin and the lengths OS and OR are equal.

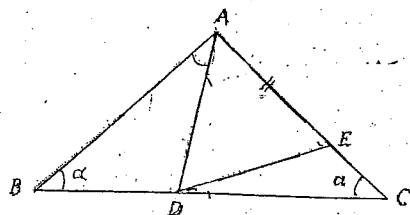


Let the length $OR = x$.

- (i) Show that the area of the rectangle is represented by the expression $A = 2xe^{-x^2}$. 1
- (ii) Find the value of x for which $PQRS$ has a maximum area. 3

Question 10 continues on page 14

Question 10 (continued)



In the isosceles triangle ABC , $\angle ABC = \angle ACB = \alpha$. The points D and E lie on BC and AC , so that $AD = AE$, as shown in the diagram. Let $\angle BAD = \beta$.

- | | | |
|-------|--|---|
| (i) | Explain why $\angle ADC = \alpha + \beta$. | 1 |
| (ii) | Find $\angle DAC$ in terms of α and β . | 2 |
| (iii) | Hence, or otherwise, find $\angle EDC$ in terms of β . | 2 |

End of Question 10

End of paper

TRIAL HSC MATHEMATICS 2011 SOLUTIONS

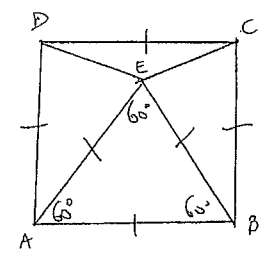
Qn	Solutions	Marks	Comments; Criteria
1 a)	$(x-3)^2 + (y+7)^2 = 25$	1	
b)	$\frac{a+b}{a-b} = \frac{\frac{1}{2} - \frac{1}{3}}{\frac{1}{2} + \frac{1}{3}}$ $= \frac{\frac{1}{6}}{\frac{5}{6}}$ $= \frac{1}{5}$	1	
c)	$8x^2 - 2x = 0$ $2x(4x-1) = 0$ $2x = 0$ or $4x-1 = 0$ $\therefore x = 0, \frac{1}{4}$	2	
d)	$\frac{\sqrt{3}}{3} + 2\sqrt{3} = \frac{\sqrt{3} + 6 - 3\sqrt{3}}{3}$ $= \frac{6 - 2\sqrt{3}}{3}$	2	
e)	$\sum_{n=1}^{100} (4n+3) = 7 + 11 + 15 + \dots + 403$ $S_{100} = \frac{100}{2} (7 + 403)$ $= 20500$	2	
f)	$\lim_{x \rightarrow 2} \frac{x-2}{(x-2)(x^2+2x+4)} = \lim_{x \rightarrow 2} \frac{1}{x^2+2x+4}$ $= \frac{1}{12}$	2	
g)	$S_{\infty} = \frac{9}{1-r}$ $48 = \frac{4}{1-r}$ $48 - 48r = 4$ $-48r = -44$ $\therefore r = \frac{11}{12}$	2	

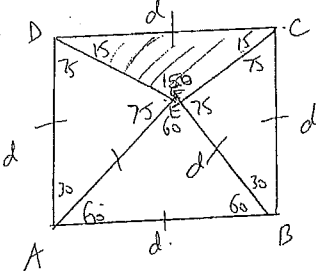
Qn	Solutions	Marks	Comments; Criteria
2 a)	$ 2x-3 > 5$ $2x-3 > 5$ or $2x-3 < -5$ $2x > 8$ or $2x < -2$ $x > 4$ or $x < -1$ $\therefore x < -1$ or $x > 4$	3	
b)	$\frac{d}{dx} x e^{\sin x} = 1 e^{\sin x} + x \cdot \cos x e^{\sin x}$ $= e^{\sin x} (1 + x \cos x), \text{ as required.}$	1	
c)	$\int \frac{1}{(5x-7)^3} dx = \int (5x-7)^{-3} dx$ $= \frac{(5x-7)^{-2}}{-2 \times 5} + C$ $= \frac{(5x-7)^{-2}}{-10} + C$ $\left(= -\frac{1}{10(5x-7)^2} + C \right)$	1	
d) (i)	$\int_0^1 e^{\pi x} dx = \left[\frac{1}{\pi} e^{\pi x} \right]_0^1$ $= \frac{1}{\pi} e^{\pi} - \frac{1}{\pi} e^0$ $= \frac{1}{\pi} (e^{\pi} - 1)$	2	
(ii)	$\int_{-1}^0 \frac{dx}{2x+3} = \frac{1}{2} \int_{-1}^0 \frac{2}{2x+3} dx$ $= \frac{1}{2} [\ln(2x+3)]_{-1}^0$ $= \frac{1}{2} [\ln 3 - \ln 1]$ $= \frac{1}{2} \ln 3$	2	

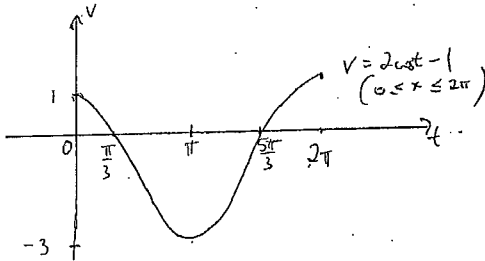
Qn	Solutions	Marks	Comments; Criteria
	Q3 continued (vi) $A = bh$. $d_{AB} = \sqrt{3^2 + (1+3)^2}$ $= \sqrt{25}$ $= 5 \text{ units}$ Area of parallelogram $= 5 \times \frac{44}{5}$ $= 44 \text{ u}^2$	2	

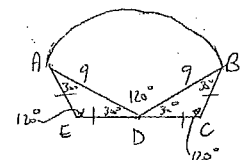
Qn	Solutions	Marks	Comments; Criteria											
4	a) (i). $T_4 = -27$ $T_7 = 729$ $\therefore ar^3 = -27 \quad \text{①}$ $ar^6 = 729 \quad \text{②}$ $\text{②} \div \text{①} \quad r^3 = -27$ $\therefore r = -3$ Sub $r = -3$ into ①: $-27a = -27$ $\therefore a = 1$ $\therefore a = 1, r = -3$ (ii) $S_n = \frac{a(1-r^n)}{1-r}$ $\therefore S_7 = \frac{1(1-(-3)^7)}{1-(-3)}$ $= \frac{2187}{4}$ $= 546.75$ b) (i) <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$\frac{4}{x(x+1)}$</td><td>2</td><td>0.67</td><td>0.33</td><td>0.2</td><td>0.13</td></tr></table> (ii) $h=1$ $\therefore A = \frac{1}{3} [(2+0.13) + 4(0.67+0.2) + 2(0.33)]$ $= 2.09$ (iii) $\frac{4}{x} - \frac{4}{x+1} = \frac{4(x+1) - 4x}{x(x+1)}$ $= \frac{4x+4-4x}{x(x+1)}$ $= \frac{4}{x(x+1)}, \text{ as required}$	x	1	2	3	4	5	$\frac{4}{x(x+1)}$	2	0.67	0.33	0.2	0.13	2 <
x	1	2	3	4	5									
$\frac{4}{x(x+1)}$	2	0.67	0.33	0.2	0.13									

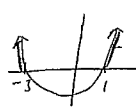
Qn	Solutions	Marks	Comments: Criteria																								
	Question 5 continued.																										
c)	$L = 120(40-t)^2$																										
(i)	$\frac{dL}{dt} = 240(40-t) \cdot -1$ $\frac{dL}{dt} = -240(40-t)$ When $t = 6$, $\frac{dL}{dt} = -240(40-6)$ $\frac{dL}{dt} = -8160 \text{ L/min}$ Water is draining out of the pool at 8160 L/min .	2																									
(ii)	$0 = 120(40-t)^2$ $\therefore (40-t)^2 = 0$ $\therefore t = 40$ $\therefore$ Pool will completely empty after 40 minutes.	1																									
d)	$\frac{dy}{dx} = x^3(x-2)(x+3)$ Let $x^3(x-2)(x+3) = 0$ $\therefore x = 0, 2, -3$ At $x = 0$ <table border="1"> <tr><td>x</td><td>-1</td><td>0</td><td>1</td></tr> <tr><td>y'</td><td>ve</td><td>-</td><td>ve</td></tr> </table> $\therefore$ At $x = 0$, a maximum turning pt exists. At $x = 2$ <table border="1"> <tr><td>x</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>y'</td><td>ve</td><td>-</td><td>ve</td></tr> </table> At $x = -3$ <table border="1"> <tr><td>x</td><td>-4</td><td>-3</td><td>-2</td></tr> <tr><td>y'</td><td>ve</td><td>-</td><td>ve</td></tr> </table>	x	-1	0	1	y'	ve	-	ve	x	1	2	3	y'	ve	-	ve	x	-4	-3	-2	y'	ve	-	ve	2	
x	-1	0	1																								
y'	ve	-	ve																								
x	1	2	3																								
y'	ve	-	ve																								
x	-4	-3	-2																								
y'	ve	-	ve																								

Qn	Solutions	Marks	Comments: Criteria
6	a) let $u = e^{2x}$ $\therefore (e^{2x})^2 + 2e^{2x} - 8 = 0$ $u^2 + 2u - 8 = 0$ $(u+4)(u-2) = 0$ $\therefore u = -4 \text{ or } 2$ $\therefore e^{2x} = -4 \text{ or } e^{2x} = 2$ No solution $2x \ln e = \ln 2$ $\therefore x = \frac{\ln 2}{2}$ $\therefore$ when is $x = \frac{\ln 2}{2}$.	3	
b)	(i) Particle is stationary at $t = 1, 5$ (ii) Velocity is negative when $1 < t < 5$ (iii) Acceleration is positive when $3 < t < 6$	1 1 1	
c) (i)	 Since $\triangle ABE$ is equilateral and $ABCD$ is a square, then $\angle EBC = 90 - 60 = 30^\circ$, as required.	1	

Qn	Solutions	Marks	Comments; Criteria
Q6 continued			
(ii)	In ΔEBC, EAD; $\angle EBC = \angle EAD$ $= 30^\circ$ (ΔEAB equilateral and $ABCD$ square) $EB = EA$ (equal sides of equilateral ΔEAB) $BC = AD$ (equal sides of square) $\therefore \Delta EBC \equiv \Delta EAD$ (SAS)	2	
(iii)	 ΔAED and ΔBEC isosceles. All angles are marked in diagram. In ΔBCE; $EC^2 = d^2 + d^2 - 2d^2 \cos 30^\circ$ $= 2d^2 - 2d^2 \cdot \frac{\sqrt{3}}{2}$ $= 2d^2 - \sqrt{3}d^2$ ΔCDE is isosceles, $EC = ED$. $\therefore$ Area of $\Delta CDE = \frac{1}{2} \cdot EC \cdot ED \sin 150^\circ$ $= \frac{1}{2} EC^2 \sin 30^\circ$ $= \frac{1}{2} (2d^2 - \sqrt{3}d^2) \cdot \frac{1}{2}$ $= \frac{d^2(2 - \sqrt{3})}{4} \text{ cm}^2,$ as required.	3	

Qn	Solutions	Marks	Comments; Criteria
7	$v = 2 \cos t - 1$ (i) $x = \int (2 \cos t - 1) dt$ $x = 2 \sin t - t + C$ At $t = 0$, $x = -1$ $\therefore -1 = 2 \sin 0 - 0 + C$ $\therefore C = -1$ $\therefore x = 2 \sin t - t - 1$	2	
(ii)	When $v = 0$, $2 \cos t - 1 = 0$ $2 \cos t = 1$ $\cos t = \frac{1}{2}$ $\therefore t = \frac{\pi}{3} \text{ s.}$ $\therefore$ Particle is first at rest when $t = \frac{\pi}{3} \text{ s.}$	2	
(iii)	When $t = \frac{\pi}{3}$, $x = 2 \sin \frac{\pi}{3} - \frac{\pi}{3} - 1$ $= 2 \cdot \frac{\sqrt{3}}{2} - \frac{\pi}{3} - 1$ $= \sqrt{3} - \frac{\pi}{3} - 1 \text{ m.}$	1	
(iv)		2	

Qn	Solutions	Marks	Comments: Criteria
	Question 7 continued (i) $k = r\theta$ $6\pi = r \cdot \frac{2\pi}{3}$ $r = \frac{18\pi}{2\pi}$ $\therefore r = 9 \text{ m (as required)}$  (ii) $\frac{BC}{\sin 30} = \frac{9}{\sin 120}$ $BC = \frac{9 \sin 30}{\sin 60}$ $= \frac{9 \cdot \frac{1}{2}}{\frac{\sqrt{3}}{2}}$ $= \frac{9}{2} \times \frac{2}{\sqrt{3}}$ $= \frac{9\sqrt{3}}{2}$ $\therefore BC = 3\sqrt{3} \text{ units}$ Note: $\sin 120 = \sin 60$ (iii). Area of pool = $\frac{1}{2} r^2 \theta + 2 \Delta's$ $= \frac{1}{2} \cdot 81 \cdot \frac{2\pi}{3} + 2 \times \frac{1}{2} (3\sqrt{3}) \sin 60$ $= 27\pi + 27 \cdot \frac{\sqrt{3}}{2} \text{ cm}^2$ $= 27 \left(\pi + \frac{\sqrt{3}}{2} \right) \text{ cm}^2$	1 2 2	

Qn	Solutions	Marks	Comments: Criteria
8	a) $b^2 - 4ac < 0$ $(-4k)^2 - 4(5k-3)(k+1) < 0$ $16k^2 - 4(5k^2 + 2k - 3) < 0$ $16k^2 - 20k^2 - 8k + 12 < 0$ $-4k^2 - 8k + 12 < 0$ $k^2 + 2k - 3 > 0$ $(k-1)(k+3) > 0$ $\therefore k < -3 \text{ or } k > 1$  (b) $\frac{3}{4}$ G $\frac{3}{4}$ G $\frac{3}{4}$ G $\frac{1}{4}$ B $\frac{1}{4}$ B $\frac{1}{4}$ B $\frac{1}{4}$ B $\frac{1}{4}$ G $\frac{1}{4}$ B $\frac{1}{4}$ B $\frac{1}{4}$ B $\frac{1}{4}$ B (i) $P(GG) = \frac{3}{4} \times \frac{3}{4}$ $= \frac{9}{16}$ (ii). $P(GGG \text{ or } GGB \text{ or } GBB \text{ or } BBB})$ $= \left(\frac{3}{4}\right)^3 + 3 \times \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)$ $= \frac{27}{32}$	3 2 2	

Qn	Solutions	Marks	Comments: Criteria
	Question 8 continued. c). (i) $y = \sin x$ ① $y = \sqrt{3} \cos x$ ② ① $\div$ ② $1 = \frac{\sin x}{\sqrt{3} \cos x}$ $\therefore \tan x = \sqrt{3}$ $\therefore x = \frac{\pi}{3}, \pi + \frac{\pi}{3}$ $= \frac{\pi}{3}, \frac{4\pi}{3}$ When $x = \frac{\pi}{3}$, $y = \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$ $\therefore$ Point A = $(\frac{\pi}{3}, \frac{\sqrt{3}}{2})$ as required. (ii). When $x = \frac{4\pi}{3}$, $y = \sin \frac{4\pi}{3} = -\frac{\sqrt{3}}{2}$ $\therefore$ Point B = $(\frac{4\pi}{3}, -\frac{\sqrt{3}}{2})$ (iii). Area = $\int_{\pi/3}^{4\pi/3} (\sin x - \sqrt{3} \cos x) dx$ $= [-\cos x - \sqrt{3} \sin x]_{\pi/3}^{4\pi/3}$ $= (-\cos \frac{4\pi}{3} - \sqrt{3} \sin \frac{4\pi}{3}) - (-\cos \frac{\pi}{3} - \sqrt{3} \sin \frac{\pi}{3})$ $= (\frac{1}{2} - \sqrt{3} \cdot (-\frac{\sqrt{3}}{2})) - (-\frac{1}{2} - \sqrt{3} \cdot \frac{\sqrt{3}}{2})$ $= \frac{1}{2} + \frac{3}{2} + \frac{1}{2} + \frac{3}{2}$ $= 4 \text{ u}^2$	1	

Qn	Solutions	Marks	Comments: Criteria
9	a) (i) $399 \times 60 = \\$23940$ $\therefore$ Interest = $\\$23940 - \\$20000 = \\$3940$ (ii). $A_1 = 20000(1+0.006)^1 - 399$ $7.27\% = \frac{7.2}{100}$ $= 0.06\% \text{ per month}$ $= 0.006$ $= \\$19721$, as required. (iii). $A_2 = A_1(1.006) - 399$ $= (20000(1.006) - 399)(1.006) - 399$ $= 20000(1.006)^2 - 399(1+1.006)$, as required. (iv) $A_n = 20000(1.006)^n - 399(1+1.006+\dots+1.006^{n-1})$ $= 20000(1.006)^n - 399 \left[\frac{1(1.006^n - 1)}{1.006 - 1} \right]$ $= 20000(1.006)^n - 399 \frac{(1.006^n - 1)}{0.006}$ $= 20000(1.006)^n - 66500(1.006^n - 1)$ $= 20000(1.006)^n - 66500(1.006)^n + 66500$ $= 66500 - 46500(1.006)^n$, as required	1 1 2 2	

Qn	Solutions	Marks	Comments: Criteria
(v)	2 years = 24 months $A_{24} = 66500 - 46500 \times 1.006^{24}$ $= \$12820.99$ $\therefore$ Mark will pay \$12820.99 to repay the loan.	1	
(b) (i)	$\phi = \phi_0 e^{-kt}$ $\frac{d\phi}{dt} = -k\phi_0 e^{-kt}$ $= -k\phi, \text{ as required.}$	1	
(ii)	$1 = 2 e^{-k \cdot 5500}$ $\frac{1}{2} = e^{-5500k}$ $\therefore \ln \frac{1}{2} = -5500k$ $\therefore k = \frac{\ln \frac{1}{2}}{-5500}$ $= \frac{\ln 2^{-1}}{-5500}$ $= \frac{-\ln 2}{-5500}$ $\therefore k = \frac{\ln 2}{5500}, \text{ as required.}$	2	
(iii)	$0.5 = 1 \cdot e^{-\frac{\ln 2}{5500} t}$ $\therefore \ln 0.5 = -\frac{\ln 2}{5500} t$ $\therefore t = \frac{\ln 0.5 \times 5500}{-\ln 2}$ $\therefore t = 15000 \text{ years (nearest 1000 yrs)}$	2	

Qn	Solutions	Marks	Comments: Criteria
(10) (a)	$\log_2 x^{-1} + \log_2 x^{-2} + \log_2 x^{-3} + \dots + \log_2 x^{-10}$ $= -\log_2 x - 2\log_2 x - 3\log_2 x - \dots - 10\log_2 x$ Series is an AP, with $a = -\log_2 x$, $S_n = \frac{n}{2}(a+l)$ $d = -\log_2 x$, $n = 10$. $440 = \frac{10}{2} [-\log_2 x - 10\log_2 x]$ $440 = 5 (-11\log_2 x)$ $88 = -11\log_2 x$ $-8 = \log_2 x$ $\therefore x = 2^{-8}$ $\therefore x = \frac{1}{256}$	3	
(b) (i)	OR $0.5 = x$ $\therefore 5x = 2x$ units Area of rectangle = 16 $= 2x \times e^{-x^2}$ $A = 2x e^{-x^2}, \text{ as required.}$	1	
(ii)	$\frac{dA}{dx} = 2x(-2x)e^{-x^2} + 2(e^{-x^2})$ $= 2e^{-x^2}(-2x^2 + 1)$ Let $\frac{dA}{dx} = 0$. i.e. $2e^{-x^2}(-2x^2 + 1) = 0$. $2e^{-x^2} = 0$ or $1 - 2x^2 = 0$ No solution $2x^2 = 1$ $x^2 = \frac{1}{2}$ $\therefore x = \pm \frac{1}{\sqrt{2}}$ But $x > 0$. $\therefore x = \frac{1}{\sqrt{2}}$		

